Progress and Prediction

Inflationary cosmology's insights into the horizon and flatness problems represent tremendous progress. For cosmological evolution to yield a homogeneous universe whose matter/energy density is even remotely close to what we observe today, the standard big bang model requires precise, unexplained, almost eerie fine-tuning of conditions early on. This tuning can be assumed, as the staunch adherent to the standard big bang advocates, but the lack of an explanation makes the theory seem artificial. To the contrary, regardless of the detailed properties of the early universe's matter/energy density, inflationary cosmological evolution *predicts* that the part we can see should be very nearly flat; that is, it *predicts* that the matter/energy density we observe should be very nearly 100 percent of the critical density.

Insensitivity to the detailed properties of the early universe is a wonderful feature of the inflationary theory, because it allows for definitive predictions irrespective of our ignorance of conditions long ago. But we must now ask: How do these predictions stand up to detailed and precise observations? Do the data support inflationary cosmology's prediction that we should observe a flat universe containing the critical density of matter/energy?

For many years the answer seemed to be "Not quite." Numerous astronomical surveys carefully measured the amount of matter/energy that could be seen in the cosmos, and the answer they came up with was about 5 percent of the critical density. This is far from the enormous or minuscule densities to which the standard big bang naturally leads—without artificial fine-tuning—and is what I alluded to earlier when I said that observations establish that the universe's matter/energy density is not thousands and thousands of times larger or smaller than the critical amount. Even so, 5 percent falls short of the 100 percent inflation predicts. But physicists have long realized that care must be exercised in evaluating the data. The astronomical surveys tallying 5 percent took account only of matter and energy that gave off light and hence could be seen with astronomers' telescopes. And for decades, even before the discovery of inflationary cosmology, there had been mounting evidence that the universe has a hefty dark side.